

Erdős Problem #1100

EP-1100

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

If $1 = d_1 < \dots < d_{\tau(n)} = n$ are the divisors of n , then let $\tau_{\perp}(n)$ count the number of i for which $(d_i, d_{i+1}) = 1$.

Is it true that $\tau_{\perp}(n)/\omega(n) \rightarrow \infty$ for almost all n ? Is it true that $\tau_{\perp}(n) < \exp((\log n)^{o(1)})$ for all n ?

Let $g(k) = \max_{\omega(n)=k} \tau_{\perp}(n)$, where $\omega(n)$ counts the number of distinct prime divisors of n , and n is restricted to squarefree integers. Determine the growth of $g(k)$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>